

EXAMINING THE ROLE OF DOPAMINE IN COMPULSIVE BEHAVIOURS OF OCD

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ABSTRACT

Can dopamine antagonists targeting D2 in the Basal Ganglia reduce compulsive behaviours in patients with OCD? This study aims to investigate how risperidone can be used as a drug intervention.

INTRODUCTION

- **Compulsions**: repetitive, maladaptive behaviours that provide varying levels of relief [1].
- **Basal Ganglia**: responsible for reward-seeking behaviour in the brain [2].
- SSRIs are not always effective for treatment-resistant OCD.
- **Risperidone**: used to augment (increase effectiveness of) SSRIs. 2mg dosage given to participants
- **PET scan**: used to confirm that reduction of D2 is correlated to reduction of symptoms [3].

DESIGN + METHOD

- 36 participants, 3 groups
1. Healthy control group
 2. Drug naive OCD patients
 3. OCD patients taking SSRIs

OCD symptom severity will be assessed using Yale-Brown Obsessive Compulsive Scale (Y-BOCS) [3] before and after course of risperidone. PET scans will be taken **before** the course commences and **8 weeks later**.

ETHICS

Participants will be warned of side effects and can reduce their dosage if needed. A **clinician** will monitor their condition to ensure they are safe. Participants must sign **consent** forms prior to the study and may **withdraw** at any time.

TARGETING DOPAMINERGIC SYSTEMS MAY REDUCE COMPULSIONS IN OCD

DISCUSSION

Exploring the role of dopamine can develop our understanding of the neuropathy of OCD and develop **alternative drug therapies**, which could optimise treatment. This would alleviate the burden on patients and the healthcare system. A potential future avenue for study is to clarify whether different subtypes of OCD are affected by different dopaminergic mechanisms or different areas of the brain to specialise treatment further.

LIMITATIONS

This study may be difficult to replicate due to how expensive PET scans are. It also only observes the initial stages of the drug intervention, not the **long-term effects**.

REFERENCES

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